

CSE331 Theory Assignment 03

1. Complete the following question:

Consider the following context-free grammar.

$$\begin{aligned} S &\rightarrow PbQ \\ P &\rightarrow aP \mid bP \mid \varepsilon \\ Q &\rightarrow aQa \mid aQb \mid bQa \mid bQb \mid \varepsilon \end{aligned}$$

- (a) Give a leftmost derivation for the string 'abaabbab'. (3 points)
- (b) Draw the parse tree corresponding to the derivation you gave in (a). (2 points)
- (c) Show that the given grammar is ambiguous by drawing two more parse trees (apart from the one that you have drawn in (b)) for the given string in (a). (4 points)
- (d) Write two strings of length four that have exactly one parse tree in the given grammar. (1 point)
- (e) Give the rightmost derivation for the string "abaabbab" and its corresponding parse tree.

2. Complete the following CFG creation question:

$$L_1 = \{ w \in \{c, d\}^* \mid \text{starts and ends with different characters} \}$$

$$A \rightarrow cX d \mid dX c$$

$$X \rightarrow 0X \mid 1X \mid \varepsilon$$

$$L_2 = \{ w \in \{0, 1\}^* \mid w \text{ is not a palindrome} \}$$

$$L_3 = \{ w \in \{0, 1\}^* \mid \text{equal number of 0's and 1's} \}$$

Similar problem $\rightarrow$ [Link](#)

$$L_4 = \{ w \in \{0, 1\}^* \mid \text{odd number of 0's and even number of 1's} \}$$

Similar problem $\rightarrow$ [Link](#)

$$L_5 = \{ w \in \{0, 1\}^* \mid \text{every 4th position is 0 and 6th position is 0} \}$$

$$A \rightarrow XXX0X0A \mid XXX0X0 \mid XXX0X \mid XXX0 \mid XXX \mid XX \mid X$$

$$X \rightarrow 0 \mid 1$$

$$L_6 = \{ w \in \{0, 1\}^* \mid \text{more 1s than 0s} \}$$

$$S \rightarrow 1S0 \mid 0S1 \mid SS \mid 1S \mid \varepsilon$$

$$L_7 = \{ w \in \{0, 1\}^* \mid \text{Starts with 101} \}$$

$$A \rightarrow 101X$$

$$X \rightarrow 0X \mid 1X \mid \varepsilon$$

$L_8 = \{ w \in \{0, 1\}^* \mid \text{Contains } 101 \}$

$A \rightarrow X101X$

$X \rightarrow 0X \mid 1X \mid e$

$L_9 = \{ w \in \{0, 1\}^* \mid \text{Does not end with } 11 \}$

$A \rightarrow X101X$

$X \rightarrow 0X \mid 1X \mid e$

$L_{10} = \{ w \in \{0, 1\}^* \mid 0^i 1^j 2^k ; k = i + j \text{ where } i, j \geq 2 \}$

$L_{11} = \{ w \in \{0, 1\}^* \mid 0^i 1^j 2^k ; i = j \text{ or } j > k \text{ where } i, j, k \geq 1 \}$

For $i = j$

$A \rightarrow BC$

$B \rightarrow 0B1 \mid 01$

$C \rightarrow 2C \mid 2$

For $j > k$

$P \rightarrow QST$

$Q \rightarrow 0Q \mid 0$

$S \rightarrow 1S \mid 1$

$T \rightarrow 1T2 \mid 12$

So combining both under union,

$S \rightarrow A \mid P$

$L_{12} = \{ w \in \{0, 1\}^* \mid 0^i 1^j 2^k ; i \neq j \text{ where } i, j, k \geq 0 \}$

Hint: One CFG for $i > j$ and another cfg for $i < j$. Then both cfg are made in union.

Number of 2 is independent. See the last language of the assignment.

$L_{13} = \{ w \in \{0, 1\}^* \mid 0^i 1^j 2^k ; k > i + 2j \text{ where } i, j, k \geq 0 \}$

$L_{14} = \{ w \in \{0, 1\}^* \mid 0^i 1^j 2^k ; i > j + k \text{ where } i, j, k \geq 1 \}$

$L_{15} = \{ w \in \{0, 1\}^* \mid w_1 \# w_2 : w_1 \in \Sigma^*, w_2 \text{ contains exactly two 1's, } |w_1| = |w_2| \}$

$A \rightarrow \underline{0A0} \mid 1A0 \mid 0B1 \mid 1B1$

$B \rightarrow 0B0 \mid 1B0 \mid 0C1 \mid 1C1$

$C \rightarrow 0C0 \mid 1C0 \mid \#$

$L_{16} = \{ w \in \{0, 1\}^* \mid \text{every } 0 \text{ is followed by even number of } 1\text{'s} \}$

$A \rightarrow 1A \mid B \mid e$

$B \rightarrow 0C$

$C \rightarrow 11C \mid 1 \mid 1A$

$L_{17} = \{ w \in \{0, 1\}^* \mid 0^n 1^{2n+3} \text{ where } n \geq 2 \}$

$S \rightarrow 00A1111111$

$A \rightarrow 0A11 \mid 111$

$L_{18} = \{ w \in \{0, 1\}^* \mid 0^n 1^m \text{ where } m \leq 2n \}$

$A \rightarrow 0A1 \mid 0A11 \mid e$

$L_{19} = \{ w \in \{0, 1\}^* \mid 0^n 1^m \text{ where } m \neq n \}$

For $m < n$

$S \rightarrow 0S \mid 0A$

$A \rightarrow 0A1 \mid e$

For $m > n$

$P \rightarrow P1 \mid Q1$

$Q \rightarrow 0A1 \mid e$

So combining both under union,

$F \rightarrow S \mid P$